

1. Unfolding and Proving.

a) $T(1) = 1$

$$T(n) = 100 \cdot T\left(\frac{n}{10}\right) + n^2.$$

We substitute $T\left(\frac{n}{10}\right)$ for its implement $\frac{n}{10}$ instead of n .

$$T\left(\frac{n}{10}\right) = 100 \cdot T\left(\frac{n}{10^2}\right) + \frac{n^2}{10^2}$$

Then substitute to the previous example

$$T(n) = 100 \cdot \left(100 \cdot T\left(\frac{n}{10^2}\right) + \frac{n^2}{10^2} \right) + n^2 =$$

$$100^2 \cdot T\left(\frac{n}{10^2}\right) + 2n^2$$

It holds for every

$$T(n) = 100^i \cdot T\left(\frac{n}{10^i}\right) + i \cdot n^2$$

We know that the recursion will stop at 1 so:

$$\frac{n}{10^i} = 1$$

$$n = 10^i$$

$$i = \log_{10} n$$

Substitute it to the initial formula to cancel i .

$T\left(\frac{n}{10^i}\right)$ is substituted into $T(1)$; i into $\log_{10} n$; 100^i into $(10^i)^2$ but $10^i = n$ so n^2 .

$$T(n) = n^2 \cdot 1 + \log_{10}(n) \cdot n^2.$$

$$T(n) = n^2 + n^2 \log_{10}(n)$$

Determine O -notation of the asymptotic time

$$T(n) = \underline{O(n^2 \log n)}.$$

Proof:

Let some $c \geq 0$ and $n \geq n_0$ so:

$$T(n) \leq c \cdot n^2 \log_{10} n$$

Checking for $\frac{n}{10}$:

$$T\left(\frac{n}{10}\right) \leq c \cdot \left(\frac{n}{10}\right)^2 \log_{10} \left(\frac{n}{10}\right)$$

And implement into the initial $T(n) = 100 \cdot T\left(\frac{n}{10}\right) + 100$

$$\begin{aligned} T(n) &\leq 100 \left(c \cdot \frac{n^2}{10^2} \cdot \log_{10} \frac{n}{10} \right) + n^2 = c \cdot n^2 \cdot (\log_{10} n - 1) - \log_{10} 10 + n^2 = c \cdot n^2 \cdot (\log_{10} n - 1) + n^2 = \\ &= c \cdot n^2 \log_{10} n - c \cdot n^2 + n^2 \end{aligned}$$

$$T(n) \leq c \cdot n^2 \log_{10} n - n^2(c-1)$$

That expression can be less or equal than $c \cdot n^2 \log_{10} n$ only if $c-1 \geq 0$, or $c \geq 1$.

we take any n_0 which is more ^{or equal} than 1 and assign for n because c doesn't make it grow.

Let it be 10 for initial:

$$T(10) = 100 \cdot T(1) + 100 = 200$$

For our assumption:

$$T(10) \leq c \cdot 10^2 \log_{10} 10$$

$$200 \leq c \cdot 100$$

Now we see $c = 2$, so it is true for $c = 2$ and $n \geq 10$.

$$6) T(1) = 1$$

$$T(n) = 3 \cdot T\left(\frac{n}{3}\right) + n.$$

$$T\left(\frac{n}{3}\right) = 3 \cdot T\left(\frac{\frac{n}{3}}{3}\right) + \frac{n}{3}.$$

$$T\left(\frac{n}{3}\right) = 3 \cdot T\left(\frac{n}{3^2}\right) + \frac{n}{3}.$$

Substitute into the initial expression:

$$T(n) = 3 \cdot \left(3 \cdot T\left(\frac{n}{3^2}\right) + \frac{n}{3}\right) + n = 3^2 \cdot T\left(\frac{n}{3^2}\right) + n + n$$

$$T(n) = 3^2 \cdot T\left(\frac{n}{3^2}\right) + 2n.$$

" " base formula:

$$T(n) = 3^i \cdot T\left(\frac{n}{3^i}\right) + i \cdot n$$

The recursion will stop at:

$$\frac{n}{3^i} = 1 \Rightarrow 3^i = n \Rightarrow i = \log_3(n)$$

Implement into the base all values:

$$T(n) = n \cdot T(1) + \log_3(n) \cdot n = n + n \log_3 n$$

$$\text{Complexity} = \underline{O(n \log n)}$$

Proof:

Let $c > 0$ and $T(n) \leq c \cdot n \log_3(n)$

Assume for the smallest $\left(\frac{n}{3}\right)$:

$$T\left(\frac{n}{3}\right) \leq c \cdot \left(\frac{n}{3}\right) \log_3\left(\frac{n}{3}\right)$$

Implement into initial $T(n)$:

$$T(n) \leq 3 \cdot \left(c \cdot \frac{n}{3} \log_3\left(\frac{n}{3}\right)\right) + n = c \cdot n \cdot (\log_3(n) - \log_3(3)) + n$$

$$T(n) \leq c \cdot n \log_3(n) - c \cdot n + n$$

$$T(n) \leq C \cdot n \log_3(n) - n(C-1)$$

For it to be less or equal to $C \cdot n \log_3(n)$ the $C-1 \geq 0$ or $C \geq 1$.

Checking:

for any $n \geq 1$; for instance 3.

$$T(3) = 3 \cdot T(1) + 3 = 3 \cdot 1 + 3 = 6.$$

Now into our assumption:

$$T(3) \leq C \cdot 3 \cdot \log_3(3)$$

$$6 \leq C \cdot 3 \cdot 1:3$$

$$C \geq 2.$$

Master theorem:

$$a) T(n) = 8 \cdot T\left(\frac{n}{2}\right) + n^3$$

$$a = 8; b = 2; f(n) = n^3.$$

To find the increasing speed finding: $\log_b a$ and $n^{\log_b a}$

$$\log_2 8 = 3 = n^3. \text{ and } f(n) = n^3.$$

They grow exactly the same, therefore 2 case is applied. So

$$T(n) = \Theta(n^3 \log n).$$

$$b) T(n) = T\left(\frac{n}{2}\right) + n \cdot \log(n)$$

$$a = 1; b = 2; f(n) = n \cdot \log n$$

$$\log_b a = \log_2 1 = 0. \quad n^0 = 1.$$

The $n=1$ It doesn't grow but stays constant.

$$\textcircled{1} \quad n=1 \\ f(n) = n \log n$$

$f(n)$ is considerably growing faster therefore case 3 is applied.

Verify additional condition:

$$a \cdot f\left(\frac{n}{b}\right) \leq c \cdot f(n) \quad (c < 1)$$

$$1 \cdot f\left(\frac{n}{2}\right) \leq c \cdot n \log n$$

$$\frac{n}{2} \log\left(\frac{n}{2}\right) \leq c \cdot n \log n$$

$$\frac{n}{2} (\log n - \log 2) \leq c \cdot n \log n \quad | : n$$

$$\frac{1}{2} (\log n - \log 2) \leq c \cdot \log n$$

$$\frac{1}{2} \log n - \frac{1}{2} \log 2 \leq c \cdot \log n \quad \text{Assuming } c = \frac{1}{2}$$

$$\frac{1}{2} \log n - \frac{1}{2} \log 2 \leq \frac{1}{2} \log n$$

$$T(n) = \underline{\underline{\Omega(n \log n)}}$$

$$C) \quad T(n) = 3 \cdot T\left(\frac{n}{3}\right) + \log(n) \\ a=3; \quad b=3; \quad f(n) = \log(n)$$

$$\log_3 3 = 1; \quad n^1 = n;$$

$n > \log(n)$ n grows way faster therefore;

$$\text{case 1 } T(n) = \underline{\underline{O(n)}}$$

$$d) T(n) = 4 \cdot T\left(\frac{n}{2}\right) + n^2$$

$$a = 4$$

$$b = 2$$

$$f(n) = n^2,$$

$$\log_2 4 = 2$$

$$g(n) = n^2$$

$$g(n) = f(n) = n^2 = n^2$$

$$T(n) = \Theta(n^2 \log n) \text{ Same speed of growing}$$